



Smart City Development and Sustainable Waste Management in Indonesian Cities: A Comparative Study of Jakarta, Surabaya, and Makassar

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Abstract: Rapid urbanization in developing countries such as Indonesia has intensified environmental challenges, including waste accumulation, pollution, and limited infrastructure capacity. Smart city initiatives have therefore been promoted as a pathway toward sustainable urban management through technology, adaptive governance, and citizen participation. However, comparative analyses of how smart city strategies contribute to environmental sustainability remain limited. This study examines the contribution of smart city development to sustainable waste management in three major Indonesian cities: Jakarta, Surabaya, and Makassar. A descriptive qualitative method was employed using a comparative literature review based on academic publications and official government documents. The analysis integrated three conceptual frameworks: the smart city framework, integrated sustainable waste management (ISWM), and ecological modernization theory (EMT). The results show that each city adopted distinct strategies reflecting its local capacities and institutional contexts. Jakarta emphasizes technological modernization through refuse-derived fuel (RDF) processing and decentralized TPS3R facilities based on the reduce–reuse–recycle principle. Surabaya focuses on community participation through waste banks, composting centers, and the Benowo waste-to-energy plant. Makassar prioritizes digital innovation through the Makassar Tidak Rantasa (MTR) reporting system, the Waste Bank Management Information System (SIMBA), and global positioning system (GPS)-based sanitation fleet management. Collectively, these approaches demonstrate that environmental sustainability in smart cities depends on the synergy among technology, policy effectiveness, and public participation. The findings indicate that there is no universal model for urban waste management; instead, effective outcomes arise from locally adapted strategies that balance technical, institutional, and social dimensions. This study provides practical insights for policymakers seeking to advance the Sustainable Development Goals (SDGs), particularly SDG 11 and SDG 12.

Keywords: Ecological modernization theory; Integrated sustainable waste management; Smart city development; Urban governance; Urban sustainability; Waste management

1 Introduction

The development of cities in the modern era is marked not only by rapid population growth, but also by increasingly complex environmental challenges, especially in countries with developing economies [1–3]. Massive urbanization is often accompanied by ecological pressures in the form of air pollution, declining water quality, waste accumulation, and increased carbon emissions [4, 5]. These environmental pressures have encouraged the adoption of technology-based development approaches, particularly the smart city concept, as a strategic means of improving urban management efficiency, quality of life, and environmental sustainability [6, 7].

The smart city concept emphasizes the use of information and communication technology (ICT) to improve public services, encourage citizen participation, and manage resources efficiently. One important pillar of the smart city framework is the smart environment, which focuses on environmental preservation through the use of technology, such as air quality sensors, digital waste management systems, environmentally friendly transportation, and smart green spaces [8, 9]. However, the effectiveness of smart environment implementation in realizing sustainable cities is still debatable, especially in developing countries facing fiscal constraints, infrastructure gaps [10, 11], limited institutional capacity [12], and social and economic inequities [13, 14].

This study selected three Indonesian cities—Jakarta, Surabaya, and Makassar—as case cities, with a primary focus on waste management within the smart environment framework. These cities were selected because they represent major Indonesian urban centers with significant waste-management pressures and different approaches to smart environment implementation. Jakarta, as the capital city, faces large waste volumes and has developed technological solutions, such as refuse-derived fuel (RDF) facilities at Bantargebang integrated waste processing site (Tempat Pengolahan Sampah Terpadu, TPST Bantargebang) [15, 16]. Surabaya is known for participatory innovations through waste banks, composting centers, and the Benowo waste-to-energy plant [17–19]. Makassar has developed digital governance initiatives, including the Makassar Tidak Rantasa (MTR) reporting system, the Waste Bank Management Information System (Sistem Informasi Manajemen Bank Sampah, SIMBA) digital waste bank system, and global positioning system (GPS)-based waste transport monitoring [20–22]. These varied approaches provide a comparative basis for understanding how Indonesian cities address waste management challenges through the smart city framework under different geographical, institutional, and governance conditions.

This study aims to analyze and compare the contribution of smart city development to environmental sustainability, with a specific focus on waste management in Jakarta, Surabaya, and Makassar. It also seeks to identify effective smart practices, obstacles encountered, and contextual factors that influence the effectiveness of waste management policies within the smart environment framework. With this focus, the study is expected to contribute conceptually and practically to the formulation of smart city development policies that are more oriented toward ecological sustainability.

Although many studies have examined smart cities in Indonesia, most previous studies remain descriptive or focus mainly on specific technical aspects, such as smart governance, digital public services [23], and technology-based urban management [24–28], or urban infrastructure development [29, 30]. These studies tend to pay limited attention to social and institutional dimensions, such as community participation, regulatory effectiveness, and infrastructure capacity, which are key issues in developing countries. In addition, comparative studies across cities within a single country, especially in developing countries such as Indonesia, are still limited. Therefore, the understanding of how smart waste management is implemented across different urban contexts is still incomplete. As such, the role of urban governance arrangements in influencing environmental sustainability outcomes within smart city initiatives remains insufficiently examined, particularly across diverse urban settings in Indonesia.

This study differs from previous research by examining not only the digitization of services but also the interaction among technology, regulation, and community participation as determining factors for the success of smart environment initiatives in developing countries. Furthermore, this study integrates three analytical frameworks: the smart city framework, integrated sustainable waste management (ISWM), and ecological modernization theory (EMT).

2 Method

This study employed a qualitative comparative approach based on library research to examine smart environment policies, programs, and practices in urban waste management across three major Indonesian cities: Jakarta, Surabaya, and Makassar. The study relied on secondary data derived from academic publications, official municipal sources, policy-related reports, institutional documents, and government program information related to smart city development and urban waste management. The selection of literature and documents was conducted purposively, emphasizing relevance to smart city development, environmental governance, urban waste management, and city-level policy implementation in Jakarta, Surabaya, and Makassar. Rather than aiming for exhaustive coverage or document quantification, this approach prioritized analytical depth and contextual richness in understanding governance dynamics across the three cities.

The analysis applied a comparative thematic approach by identifying and categorizing key patterns related to technological innovation, regulatory frameworks, institutional capacity, and community participation. Relevant information was extracted from the selected sources, including city coverage, waste management programs, technological applications, policy instruments, community participation mechanisms, and reported outcomes. These findings were then compared across the three cities to identify similarities, differences, and city-specific waste management pathways.

The identified themes were mapped into the three dimensions of ISWM: technical-operational, institutional, and socio-participatory. They were further interpreted using the smart city framework and EMT to explain how technological innovation, policy effectiveness, and social participation interact in shaping urban waste management practices. To ensure analytical rigor, the study employs source triangulation by comparing insights across official municipal sources, policy-related reports, and institutional documents. Peer debriefing was also conducted through academic discussions to refine interpretation, reduce researcher bias, and maintain consistency in the analytical process. The characteristics of the data sources used in this study, including their analytical relevance and city coverage, are summarized in Table 1.

Table 1. Characteristics and representative references of data sources

Source Type	Description	Analytical Relevance	City Coverage	Representative Sources
Academic literature	Scholarly articles on smart city development, environmental governance, integrated sustainable waste management (ISWM), and ecological modernization theory (EMT), and urban waste management	Provides conceptual frameworks, theoretical support, and empirical insights	Jakarta, Surabaya, Makassar	[6–14, 20, 23–44]
Official municipal sources	City government webpages and official program information related to waste management and smart city initiatives	Provides updated information on local programs, waste-management practices, and smart city evaluation	Jakarta, Surabaya, Makassar	[16, 17, 21, 22, 45–49]
Policy-related and institutional sources	Roadmap information, institutional reports, theses, and policy-related documents	Supports contextual interpretation of city-level waste management strategies and policy direction	Jakarta, Surabaya, Makassar	[15, 18, 19, 50–52]

3 Results

3.1 Surabaya

By 2023, approximately 1,600 t/day of waste entered the Benowo final disposal site (Tempat Pemrosesan Akhir, TPA Benowo), around 60% of which was organic waste [17]. A previous study also reported that plastic accounted for 17.8% of the waste composition in Surabaya [34]. This condition places considerable pressure on TPA Benowo and highlights the importance of source-based waste reduction strategies. This situation increases the risk of air, soil, and water pollution, while emphasizing the importance of strategies that emphasize prevention from the source. Thus, the Surabaya City Government has adopted a waste management approach that combines technological innovation, regulatory measures, and community participation, positioning the city as a useful reference for urban waste management practices in Indonesia.

One of the main pillars of Surabaya's waste management system is its network of waste banks. According to an official Surabaya City Government document, the city's waste reduction activities are supported by 3 central waste banks and 670 waste bank units distributed across the city [19]. Waste banks allow residents to deposit recyclable inorganic waste with economic value, thereby reducing residual waste at temporary storage sites while providing financial benefits to participants [32–34]. The waste bank model encourages behavioral change among residents and strengthens collective awareness that waste management is a shared responsibility rather than solely a government function. For example, the Rukmi Waste Bank is reported to collect approximately 443 kg of sorted waste per month, part of which is sent to the Surabaya Central Waste Bank after exceeding the required collection threshold [31]. It also conducts reuse and recycling activities, such as producing compost, eco-enzymes, and candles from used cooking oil [31]. Beyond reducing the volume of waste sent to disposal facilities, these activities contribute to the development of a circular economy at the community level.

Evaluations of the waste bank program in Surabaya show varied results across different areas. In areas with strong community leadership and institutional support, such as the neighborhood association (Rukun Warga, RW) in Jambangan, citizen participation is relatively high and includes decision-making, implementation, and utilization of program outcomes, although formal evaluation remains limited [35]. In contrast, the Central Waste Bank shows relatively low operational effectiveness, estimated at approximately 25%, despite relatively good efficiency, which has been attributed to the limited number of daily active customers [36]. The Green and Clean model, with community involvement at the village level such as in Ngagelrejo, has helped expand participation more significantly [37]. Taken together, these initiatives reflect a gradual shift in community behavior from conventional waste disposal toward sorting, recycling, and processing practices, while also maintaining the cleanliness of the urban environment [38].

Another important program is the Jambangan Compost House and waste processing sites based on reduce, reuse, and recycle principles (TPS3R), which have shown measurable contributions to local organic waste reduction. The Jambangan facility processes around 6–7 t/day of organic waste and can reduce up to 50% of the waste that would otherwise be disposed of in landfills [53]. If similar facilities are replicated in more locations, they may contribute

substantially to reducing organic waste at the city level. Besides, the composting house also serves as an environmental education center for local residents, thereby supporting collective behavioral change. Combining technical and social functions, composting house and TPS3R facilities represent important examples of community-based waste management models.

The Jambangan recycling center (Pusat Daur Ulang, PDU Jambangan) and the Benowo waste-to-energy facility (Pembangkit Listrik Tenaga Sampah, PLTSa Benowo) are also important milestones in Surabaya's waste management. PDU Jambangan integrates various approaches, including TPS3R, composting, black soldier fly (BSF) maggot cultivation, and the use of waste for energy. With these approaches, the volume of waste entering the Benowo landfill can be reduced substantially, thereby extending the operational life of the landfill. PLTSa Benowo is reported to have an installed electricity generation capacity of 11 MW, comprising 2 MW from the landfill gas power plant and 9 MW from the gasification power plant [18]. Together, these facilities illustrate Surabaya's efforts to integrate resource recovery and renewable energy generation into urban waste management.

In addition to technical programs, the Surabaya City Government has also strengthened the regulatory dimension through Mayor of Surabaya Regulation No. 16 of 2022 concerning the reduction of plastic bag use in Surabaya City [17, 51]. This policy has succeeded in reducing plastic waste by around 1.5–2 t/day, especially in modern retail outlets. Although this achievement is encouraging, plastic waste continues to constitute a significant share of daily waste, suggesting that implementation efforts need to be expanded to traditional markets and other informal consumption spaces. The experience of Surabaya demonstrates that regulations are more effective when supported by consistent monitoring, public education, and enforcement mechanisms.

Digitalization efforts in waste management are further supported by the Solid Waste Application Transportation (SWAT) system, which supports the online monitoring and verification of waste transportation activities. Previous studies describe SWAT as a technology-based system used to monitor waste vehicles, verify waste transport data, and record waste disposal activities at TPA Benowo through barcode scanning and online data verification [52, 54, 55]. The system contributes to greater transparency and administrative control in waste transportation, although its effectiveness depends on technical capacity, staff training, and system maintenance.

When viewed through the ISWM framework, Surabaya demonstrates relatively strong interaction among technical-operational, institutional, and socio-participatory dimensions. The technical-operational dimension is reflected in PLTSa Benowo, PDU Jambangan, composting houses, and TPS3R facilities. The institutional aspect is supported by regulations on single-use plastics and the formal management of the Benowo landfill. Meanwhile, socio-participatory dimension is in the extensive network of waste banks and community-based composting initiatives. Surabaya's waste management system combines technical, regulatory, and community-based approaches rather than relying solely on technical interventions.

Within the smart environment framework, Surabaya's programs demonstrate the integration of technology, policy, and community participation. SWAT represents digital innovation for improving waste transportation efficiency, while PDU Jambangan and PLTSa Benowo illustrate the use of modern technology to reduce residual waste and generate energy. Waste banks and composting houses further demonstrate that citizen participation remains central to sustainable smart city implementation. From the perspective of EMT, Surabaya shows how waste can be repositioned from an environmental burden into a resource for energy, fertilizer, and value-added products. Multi-stakeholder collaboration among the city government, communities, and private actors also reflects a shift toward more adaptive environmental governance. Therefore, Surabaya represents a community-based circular waste management pathway supported by technological and regulatory innovation, contributing to Sustainable Development Goal (SDG) 11 (Sustainable Cities and Communities) and SDG 12 (Responsible Consumption and Production).

3.2 Jakarta

Waste management in Jakarta faces major challenges, as daily waste generation reaches approximately 7,734 t/day, with about 60% originating from residential areas [15]. This volume makes Jakarta one of the cities with the largest waste burdens in Indonesia. Therefore, waste management cannot rely solely on the Bantargebang landfill but requires an integrated system beginning at the household level. Recognizing this situation, the Special Capital Region of Jakarta (Daerah Khusus Ibukota Jakarta, DKI Jakarta) Provincial Government prepared the 2025–2026 Roadmap for the Acceleration of Waste Management Completion in Jakarta [45]. The roadmap adopts an upstream, midstream, and downstream approach, including community participation in waste sorting, waste banks, organic waste processing, TPS/TPS3R optimization, transport efficiency, RDF Plant development, and partnerships with the private sector [45]. Jakarta is gradually shifting its waste management paradigm from disposal-oriented practices toward a more sustainable circular system. The three main programs examined in this section are the Bantargebang RDF facility, the RW waste bank network, and TPS3R facilities at the sub-district level.

One of Jakarta's main technological innovations is the implementation of RDF processing at TPST Bantargebang [39]. RDF is an alternative fuel derived from processed waste and can be used as a partial substitute for coal, particularly in cement factories [40]. The facility reduces the volume of waste accumulated at the landfill while also generating

economic value through the sale of RDF to industrial users.

The Bantargebang RDF facility operates at a capacity of about 2,000 t/day, consisting of 1,000 tons of newly generated waste and 1,000 tons of waste obtained from landfill mining [16]. From this processing capacity, around 700–750 tons of RDF can be produced per day, with a calorific value comparable to low-rank coal. RDF products have been sold to the cement industry at a price of around 359,000 Indonesian rupiah (IDR) per ton. Beyond reducing the volume of waste disposed of at the landfill, the facility creates economic value by converting waste into an alternative industrial fuel. These characteristics position the Bantargebang RDF facility as an important component of Jakarta's circular economy strategy.

The Bantargebang RDF facility indicates positive contributions in technical, economic, and environmental dimensions. Technically, RDF can help reduce pressure on the Bantargebang landfill. A study on RDF processing at TPST Bantargebang reported that RDF can produce fuel with a calorific value of approximately 3,000–6,000 kcal/kg, indicating its potential as an alternative fuel, particularly for the cement industry [39]. Economically, RDF sales provide a potential revenue stream for waste management. Environmentally, RDF helps decrease the volume of waste disposed of at the landfill and reduce reliance on coal as a fossil fuel. Thereby, RDF represents one of the main technological initiatives within Jakarta's smart environment strategy.

Beyond the technological aspect, Jakarta also strengthens the social dimension through the RW waste bank network. As of December 2024, 1,878 of Jakarta's 2,748 RWs had waste banks, with 2,287 waste banks in total [48]. Of these, 1,435 waste banks, or about 63%, were active, while 852 were inactive. Waste banks play a role in encouraging community participation in 3R (Reduce, Reuse, Recycle) activities and forming a circular economy at the household level. As a result, inorganic waste volumes at TPS decrease and the community may obtain additional economic benefits [48].

Evaluations of waste banks show relatively broad geographical coverage, although their effectiveness remains uneven. In areas with strong community leadership, waste banks have contributed to reducing inorganic waste volumes and generating additional income for residents. However, in some neighborhoods, community participation is still low, limiting the overall benefits of the program. From a smart environment perspective, waste banks represent a combination of social participation and circular economy practice within the smart city ecosystem. This situation underscores the importance of more intensive strategies to strengthen the social and institutional aspects of waste management.

The third program is TPS3R, which functions as a decentralized waste processing facility. According to the Jakarta Capital City Government presentation [15], TPS3R facilities are reported in 19 locations, with a processing capacity of approximately 5–50 t/day of waste. It also notes the planned conversion of TPS to TPS3R in 13 locations, with a capacity of 25–100 t/day, and a target of expanding TPS3R development across 44 sub-districts. Given this target and the upper reported processing capacity of 50 t/day per TPS3R unit, the estimated potential waste reduction could reach approximately 2,200 t/day. TPS3R facilities reduce the amount of waste transported to Bantargebang, particularly in areas with active facilities, and strengthens community involvement at the local level.

TPS3R implementation shows that the most significant contributions are found in areas with active facilities. These facilities not only reduce waste but also serve as centers for environmental education. Within the ISWM framework, TPS3R strengthens technical-operational performance and improves logistical efficiency. However, its coverage is still limited, meaning that its overall impact is still partial. To achieve broader impact, TPS3R facilities need to be expanded to cover more sub-districts in Jakarta. Previous studies on Jakarta's household solid waste management have emphasized the potential of communal composting and source-based organic waste treatment in reducing the amount of waste sent to landfills [43].

Jakarta is relatively strong in technical-operational aspects (RDF and TPS3R facilities) and institutional aspects (waste bank development and policy support). However, community participation remains a challenge, as only about 63% of waste banks are active. As a result, the long-term success is highly dependent on social participation, public education, and consistent institutional support. A core principle of ISWM is the balance between technical, institutional, and social dimensions in achieving a sustainable waste management system.

Within the smart environment perspective, Jakarta's three main programs demonstrate the close connection between technology, society, and urban governance. The Bantargebang RDF represents the use of technology to convert waste into alternative fuel while reducing the burden on landfills. The RW waste bank reflects community empowerment and circular economy practices at the household level. Meanwhile, TPS3R emphasizes a decentralization strategy that can reduce residue and increases waste transportation efficiency. The synergy of these programs reflects that Jakarta is not only expanding infrastructure but also integrating environmental, economic, and social aspects into a more sustainability-oriented smart city system.

Viewed through the lens of EMT, the RDF program demonstrates ecological industrialization, in which waste is transformed into an alternative fuel with economic value. TPS3R facilities and waste banks reflect ecological modernization at the community level by integrating residents into recycling and circular economy practices. The collaboration among the government, cement industry, and community also illustrates the importance of multi-actor

participation in environmental management. Thus, Jakarta is modernizing its waste management system while also creating new economic value based on sustainability.

Jakarta has demonstrated measurable progress in waste management through RDF development, TPS3R expansion, and community-based waste bank programs, although challenges exist in participation scale and operational consistency. The Bantargebang RDF facility contributes to reducing landfill pressure through waste-to-fuel conversion, TPS3R facilities support decentralized waste processing at the sub-district level, and waste banks encourage community involvement in recycling and circular economy practices. The integration of smart environment principles, ISWM, and EMT has supported the transition toward a more modern, participatory, and circular-oriented waste management system in Jakarta. These developments also support the broader objectives of SDG 11 and SDG 12.

3.3 Makassar

Waste management is a crucial issue in major Indonesian cities, including Makassar, which faces environmental pressures due to population growth and economic activity. Within the smart city framework, the smart environment pillar emphasizes the use of digital technology, adaptive governance, and community participation to reduce urban ecological pressures. Makassar had implemented three main programs in waste management: the MTR program, the SIMBA digital waste bank information system, and GPS-based monitoring for the waste transportation fleet [20, 22]. These initiatives are intended to accelerate the detection of sanitation problems, expand citizen participation in the circular economy, and improve transportation logistics efficiency. With a focus on real-time data, community involvement, and operational strengthening, this policy direction is consistent with the sustainability-oriented principles of the smart environment.

The MTR program provides a digital reporting mechanism that allows citizens to report waste accumulation and pollution hotspots. This system shortens the bureaucratic chain by directly connecting community reports to the relevant agencies. By utilizing digital channels, MTR strengthens two-way communication between the government and citizens and marks a shift in environmental monitoring toward data-based governance. In a broader smart governance context, Makassar's Smart City Index score increased from 3.14 in 2023 to 3.64 in 2024, according to an official statement by the Makassar City Government [21].

The SIMBA program emphasizes social participation supported by digitalization. According to the Makassar City Environmental Agency (DLH Makassar), the Makassar City Government launched the SIMBA-Bank Sampah application to connect residents with waste bank units (BSUs) across the city. The application provides information on waste bank data, waste pick-up services, and a marketplace for recycled products [22]. With this approach, waste banks are no longer only local initiatives but are increasingly integrated into a digitalized and more organized city-wide system.

According to official data released by the Makassar City Government in July 2024, there were 1,076 productive waste banks in Makassar [46]. DLH Makassar also reported waste bank product weighing activities as part of early-year evaluation and reporting. These activities were intended to measure economically valuable waste management outputs and support the sustainability of the city's waste bank program [49]. Previous research on the Makassar City Central Waste Bank further shows that waste banks can contribute to economic and environmental sustainability, although challenges remain in terms of community participation, managerial capacity, and supporting facilities [41]. Taken together, these data and findings indicate continued citizen involvement in waste sorting and recycling while also creating local economic value. Economic incentives may strengthen community motivation to adopt pro-environmental behaviors consistently. SIMBA and the waste bank program reflect the integration of social, economic, and ecological dimensions within the framework of a smart environment.

Technical-operational aspects are strengthened through the implementation of GPS-based monitoring for the waste collection fleet. Previous research identifies GPS tracking as one of Makassar's smart city innovations in waste management, particularly for supporting the optimization of waste transport fleet operations [20]. Beyond the Makassar case, studies on municipal solid waste collection route optimization show that dynamic routing can improve collection efficiency and reduce environmental impacts by considering transport costs and greenhouse gas emissions in routing models [44].

When analyzed through the ISWM framework, Makassar shows efforts to integrate three main dimensions. GPS tracking strengthens the technical-operational aspect by improving logistical efficiency. MTR reinforces the institutional aspect through a digital reporting system that strengthens inter-agency coordination. SIMBA articulates the socio-participatory aspect by mobilizing the community in recycling and the circular economy. However, the quality of participation across regions still needs to be strengthened so that the positive impact of the program can be felt more widely.

Within the framework of a smart environment, these three programs demonstrate the close connection between technology, society, and urban governance. MTR accelerates the detection of sanitation issues through digital channels for citizens. SIMBA strengthens the culture of sorting and community-based circular economy. GPS tracking improves technical efficiency while reducing emissions from waste transportation. Makassar demonstrates ongoing

progress toward the development of a more sustainable and digitally integrated urban waste management system.

Viewed through the lens of EMT, Makassar demonstrates a process of ecological modernization that is manifested through institutional, social, and technical innovations. MTR reflects a digital governance mechanism for environmental reporting, SIMBA supports the transformation of recyclable waste into economic resources, and GPS tracking represents technical modernization in waste transportation management. Makassar's case demonstrates the role of multi-stakeholder collaboration among the government, community, and private sector in realizing an environmentally resilient city. These initiatives reflect how the smart environment approach can support urban sustainability and contribute to the achievement of SDG 11 and SDG 12.

4 Discussion

Waste management as part of smart environment implementation shows varying approaches in Jakarta, Surabaya, and Makassar. These differences can be interpreted through the combined lens of the smart city framework, ISWM, and EMT, which together explain how technological innovation, institutional capacity, and citizen participation shape waste governance in different urban contexts. More importantly, the three cities represent distinct pathways of urban waste governance, reflecting different priorities in technological development, institutional capacity, and citizen engagement within the smart city transition.

Jakarta stands out as the city with the largest waste challenge, generating approximately 7,734 t/day of waste. The approach taken is relatively consistent with EMT, where environmental modernization is carried out through RDF technology at TPST Bantargebang and collaboration with the cement industry. Jakarta represents a technological-industrial pathway, in which waste management is driven primarily by large-scale infrastructure, resource recovery, and industrial integration. However, from an ISWM perspective, Jakarta remains relatively weak in terms of social participation, as reflected in the limited activity of waste banks. This suggests that technological modernization alone is insufficient without broader community engagement, consistent with previous studies on waste governance in developing countries that emphasize the importance of socio-institutional capacity alongside technological solutions [11, 12, 42].

Surabaya presents a contrasting model through the integration of technology, regulation, and community participation. The presence of an extensive network of waste bank units and decentralized facilities such as TPS3R and composting centers reflects strong community involvement in waste reduction. These initiatives are complemented by regulatory measures and technological innovations, including waste-to-energy systems and transportation management tools. Compared to Jakarta, Surabaya reflects a community-based circular pathway, where citizen participation plays a central role alongside institutional and technological support. From an ISWM perspective, Surabaya shows a relatively balanced integration of technical, institutional, and socio-participatory dimensions. In EMT terms, ecological transformation is visible in the conversion of waste into energy and economic value. This is consistent with previous studies that identify community-based waste management and circular economy practices as key drivers of urban sustainability [34, 35, 38]. This configuration represents a more holistic model of smart environment implementation, in which multiple governance dimensions operate simultaneously.

Makassar adopts a different strategy by focusing on digitalization through applications such as MTR, the SIMBA digital waste bank system, and GPS-based fleet management. These initiatives emphasize digital reporting, information management, operational monitoring, and citizen interaction within the smart city framework [20, 22]. In contrast to Jakarta and Surabaya, Makassar represents a digital governance pathway, where technology is primarily used to enhance communication, monitoring, and operational efficiency rather than large-scale processing or community-based circular systems. However, uneven digital literacy, infrastructure readiness, and participation capacity may constrain broader implementation. From an ISWM perspective, the socio-participatory dimension remains underdeveloped, while from an EMT perspective, ecological modernization is still at an early stage. This implies that digital innovation alone does not automatically generate environmental outcomes unless it is supported by institutional capacity and sustained public participation.

In comparison, the three cities show differentiated strategies aligned with the theoretical frameworks used. Jakarta reflects EMT through its industrial and technological orientation, Surabaya aligns with ISWM through its balanced and participatory approach, and Makassar illustrates the smart city framework through digital governance. These distinctions show that smart environment development is context-dependent: Jakarta prioritizes technological efficiency, Surabaya emphasizes participatory sustainability, and Makassar focuses on digital coordination. This pattern is consistent with broader studies on smart city governance in developing countries, which emphasize that smart city implementation is shaped by institutional capacity, infrastructure readiness, governance arrangements, and socio-economic conditions [10–12]. The comparison suggests that there is no universal model for urban waste management in developing countries; instead, governance pathways are shaped by local institutional capacity, infrastructure availability, and socio-economic conditions.

Globally, these findings contribute to the literature on smart city implementation in the Global South. Previous studies have emphasized technological innovation and smart infrastructure as important components of sustainable

smart city development [8, 9]. However, this study shows that in the Indonesian context, effective waste management is not solely dependent on technological sophistication but also on regulatory frameworks and community participation. This aligns with the literature on ISWM, which emphasizes cross-city comparison, institutional soundness, inclusivity, and proactive policies [56], as well as technology-supported civic engagement in environmental sustainability [57]. Thus, Jakarta, Surabaya, and Makassar demonstrate that smart environments can emerge through diverse, locally adapted pathways rather than through a single standardized model.

A practical implication of this comparison is that smart waste management policies should be designed according to local governance capacity, infrastructure readiness, and community participation levels. Jakarta needs to strengthen social participation to complement its technological infrastructure; Surabaya needs to maintain the integration of community-based practices with regulatory and technological support; and Makassar needs to improve digital inclusion and institutional coordination to expand the impact of its smart governance initiatives. These pathways indicate that achieving SDG 11 and SDG 12 requires the alignment of technological innovation, governance capacity, and community participation.

5 Conclusion

This study shows that waste management as part of the smart environment in Jakarta, Surabaya, and Makassar follows different but complementary transition paths. Jakarta focuses on technical innovation through RDF and decentralized TPS3R facilities, which is consistent with the ecological modernization approach. Surabaya stands out for its strong community participation through waste banks, composting houses, PDU Jambangan, and PLTSa Benowo, reflecting a relatively balanced application of ISWM principles. Meanwhile, Makassar has developed a digital governance pathway through the MTR, SIMBA, and GPS tracking, demonstrating the role of digital systems in the smart city framework.

A common finding across the three cities is that the effectiveness of smart environment initiatives is not determined by a single universal model. Instead, it depends on the alignment of technology, adaptive regulation, institutional capacity, and social participation according to local urban conditions. The study contributes to offering a comparative understanding of smart waste management in the Global South by integrating the smart city framework, ISWM, and EMT. By linking these frameworks to the cases of Jakarta, Surabaya, and Makassar, the study also strengthens the relevance of smart waste governance to SDG 11 and SDG 12.

Despite its theoretical and practical contributions, this research has several limitations. Because it relies on secondary data and literature-based analysis, it does not directly capture citizen experiences, policy implementation dynamics, or micro-level variations in participation across neighborhoods. This limitation constrains the analysis of behavior and implementation variations within each city. Future research should combine qualitative methods (e.g. in-depth interviews and field observations) with quantitative approaches (e.g. participation surveys, and waste reduction or emission data analysis). Such mixed-method research approaches may provide assess the real impacts of smart environment programs more accurately, generate more operational policy recommendations, and expand the contribution of smart city literature in Southeast Asia and other developing regions.

Author Contributions

Conceptualization, I.T.S.; methodology, I.T.S. and D.B.C.; literature search and document screening, D.B.C. and E.M.N.A.; information extraction and organization, D.B.C.; comparative thematic analysis, I.T.S. and D.B.C.; validation, I.T.S., D.B.C., and E.M.N.A.; writing—original draft preparation, I.T.S. and D.B.C.; writing—review and editing, I.T.S. and E.M.N.A.; supervision, I.T.S.; project administration, I.T.S. All authors have read and agreed to the published version of the manuscript.

Data Availability

No new datasets were generated during this study. The data and information analyzed in this article were obtained from the published sources, official documents, and institutional materials cited in the manuscript.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

The authors used ChatGPT (OpenAI) solely for language editing and improving the clarity, readability, and grammar of the manuscript. The authors reviewed and edited all AI-assisted outputs and take full responsibility for the content of the manuscript. No AI tool was used for data analysis, interpretation of results, or generation of scientific conclusions.

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